CODTECH IT SOLUTIONS

Online internship program 2025

Task 2

Task 2:- Temperature monitoring System

Introduction

The Internet of Things (IoT) has revolutionized various industries, enabling the development of smart and interconnected systems. In this project, an IoT-based temperature monitoring system using the ESP32 microcontroller and Blynk app was implemented.

The system utilizes DTH11 and DS18B20 sensors to measure temperature and a 16×2 LCD display to display the temperature readings. The Blynk app provides remote monitoring and control of the system. This project demonstrates the integration of hardware, software, and cloud-based applications to create an efficient temperature monitoring solution.

Components

Hardware part

1) ESP-32 development board
2) DHT11 sensor
3) DS18B20 sensor
4) 4.7Kohm Resister
5) 16x2 LCD Display With I2C
6) Breadboard

Software part

1) Arduino IDE with ESP Boards
2) Blynk IoT Application (can be downloaded from Play-stores

Working

ESP32 Board

ESP32 is a popular microcontroller board that is based on the ESP32 system-on-chip (SoC) developed by Espressif Systems.

The ESP32 board includes a dual-core processor, built-in Wi-Fi and Bluetooth connectivity, making it suitable for a wide range of projects.

Microcontroller- 32-bit Xtensa LX6 dual-core processor, which operates at a clock speed of up to 240 MHz. The dual-core architecture allows for multitasking tasks.

Features

1)Memory: 520Kb SRAM

2)WiFi: 802.11/g/n/e/i

3)Bluetooth: v4.2 BR/EDR And BLE

4)CPU: Xtensa Dual-core(or Single-core) 32-bit LX6 Microprocessor, Operating at 160 or 240 MHz And Performing at up to 600 DMIPS

5)10 × touch sensors

6)Hall sensor

7)LED PWM up to 16 channels

8)Ultra-low Power (ULP) Co-Processor

9)12-Bit SAR ADC up to 18 channel

10)2×8-bit DAC's

11)4x SPI

12)2xI2S Interfaces

13)2xI2C Interfaces

14)3xUART

15)Low Power Mode

16×2 LCD Display

16×2 LCD display with I2C refers to a liquid crystal display (LCD) module that consists of 16 columns and 2 rows of characters.

I2C Interface- The I2C interface on the LCD display module allows you to control the display using I2C commands.

I2C Module- PCF8574 or PCF8574A acts as an I/O expander.

VCC and GND: Connect 5v power supply and GND to GND pins of the LCD display and the microcontroller.

SDA (Serial Data) and SCL (Serial Clock): I2C PIN like if used then connected the A5 and A4.

DHT11 Sensor

DHT11 sensor is a widely used digital temperature and humidity sensor in various projects. The sensor is small in size and can be easily integrated into any Microcontroller

The DHT11 sensor contains a humidity-sensing component and a temperature-sensing component.

Humidity Sensing

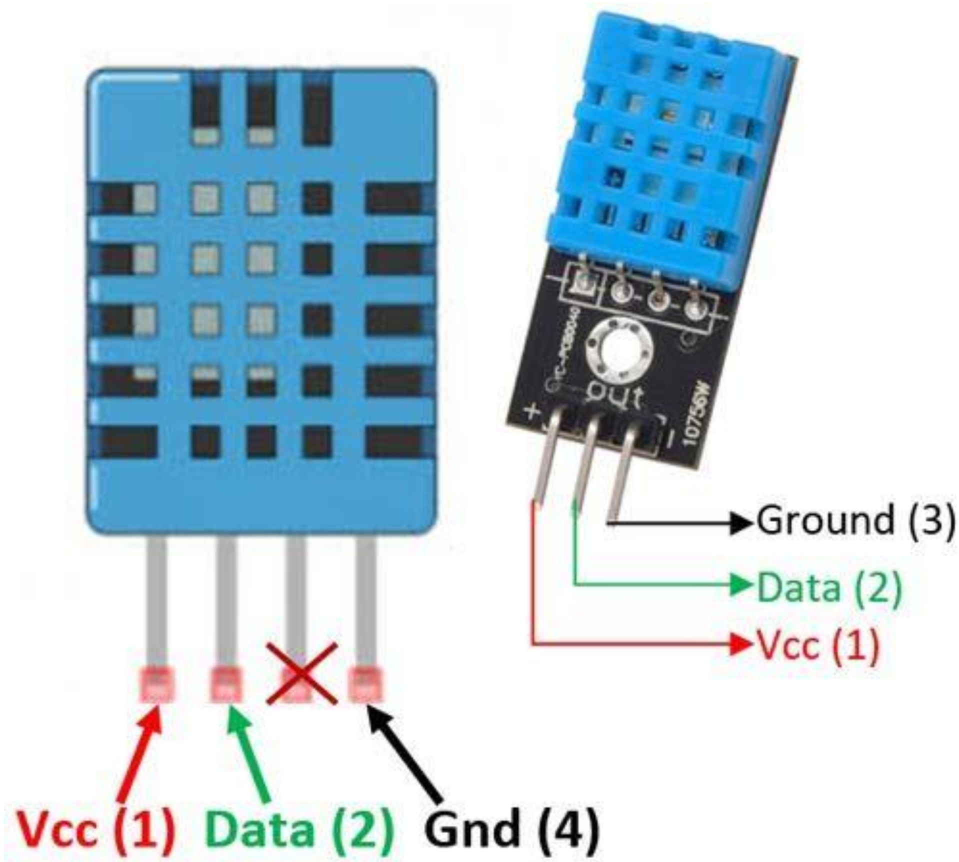
The humidity-sensing component consists of a moisture-sensitive capacitor. It absorbs moisture from the surrounding environment, causing changes in its capacitance.

Temperature Sensing

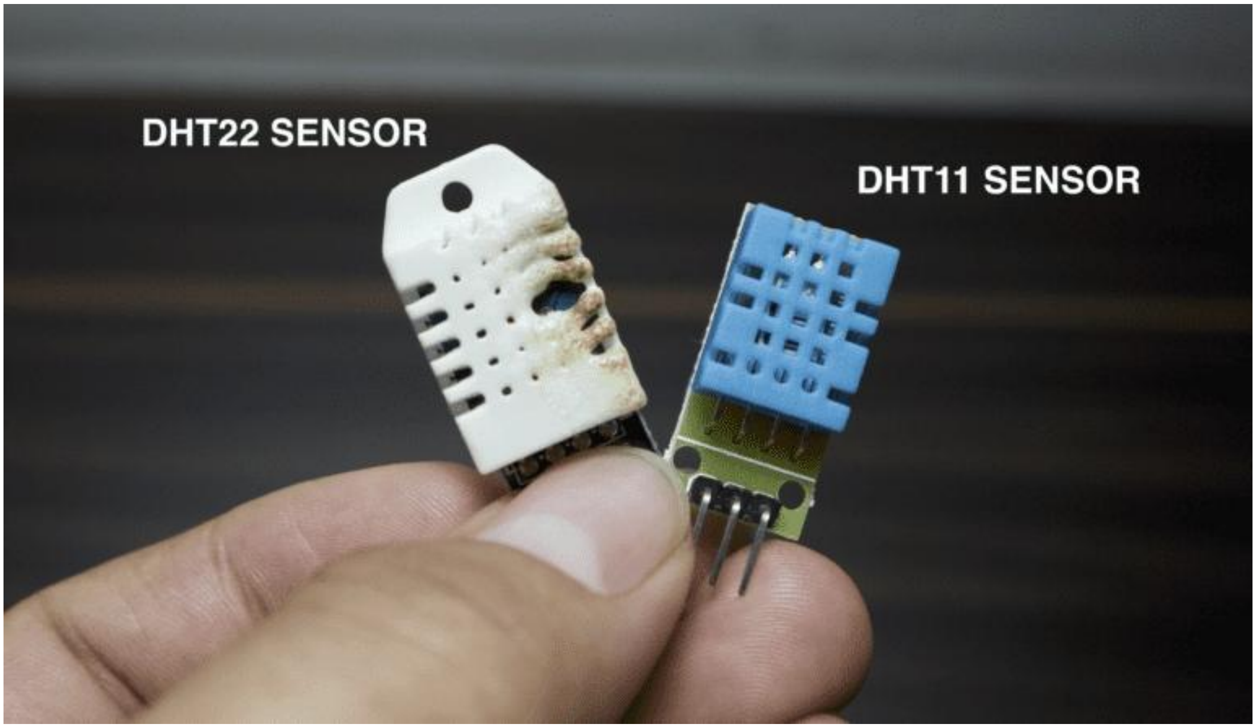
The temperature-sensing component uses a thermistor, which is a type of resistor whose resistance changes with temperature. By measuring the resistance, the sensor determines the temperature.

Analog-to-Digital Conversion

The DHT11 sensor internally performs analogue-to-digital conversion to convert the measured analog



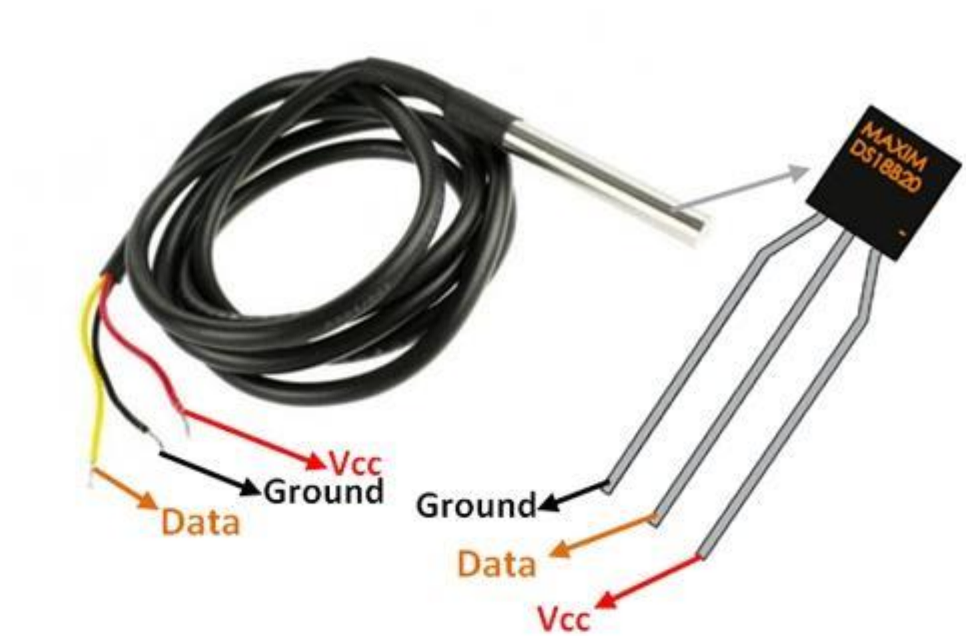
signals (humidity and temperature) into digital values that can be read by a microcontroller.



DS18B20 sensor

DS18B20 is a popular digital temperature sensor widely used in various projects and applications.

The DS18B20 uses the 1-Wire protocol, which enables communication and power supply over a single data line.



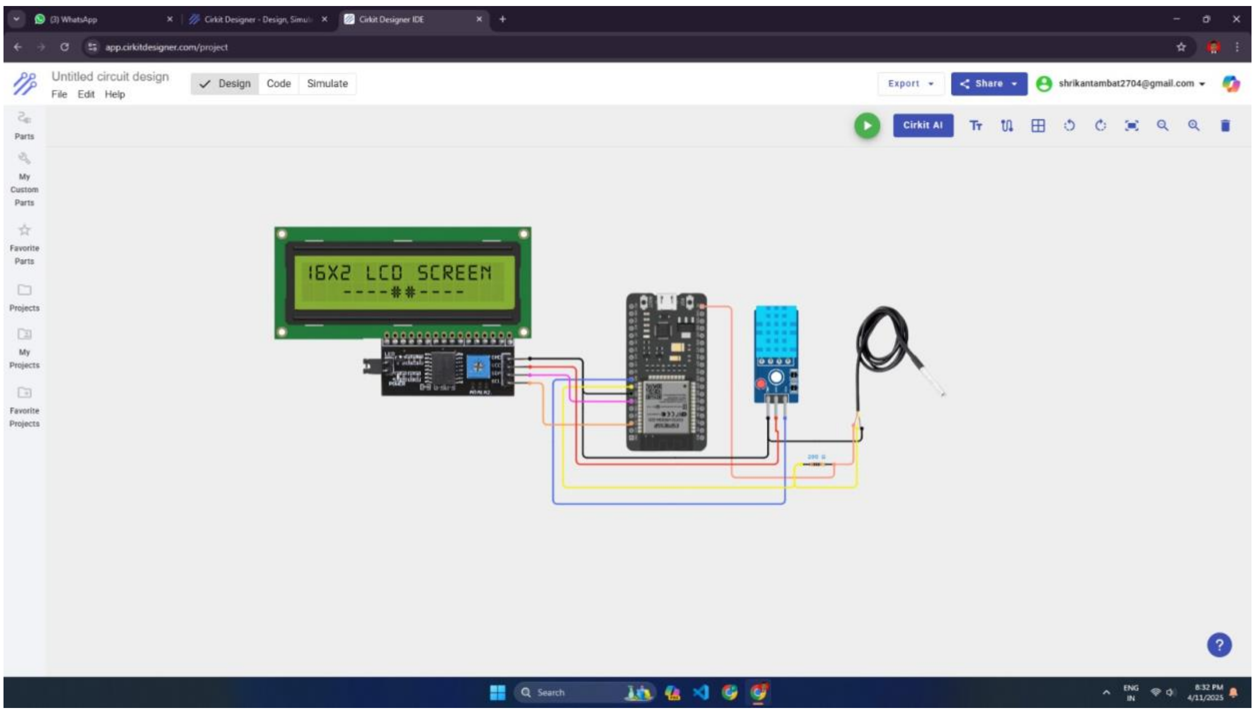
Digital Thermometer

The DS18B20 is a digital thermometer specifically designed to measure temperature. It eliminates the need for analogue-to-digital conversion by providing temperature readings directly in a digital format.

One-Wire Protocol

The DS18B20 sensor uses the 1-Wire protocol, which allows multiple sensors connected to a single data line. This connects multiple sensors to a single microcontroller pin, using a unique address for each sensor.

Circuit design



Here is an explanation of the circuit diagram for the IoT-based temperature monitoring system using ESP32, DTH11, DS18B20 sensors, and a 16×2 LCD display.

16×2 LCD is connected to the I2C protocol is connected to SDA Pin to D21 And SCL Pin connected to the D22 Pin in ESP32 Board.VCC Coneted to the 5v And GND will Connected to the GND.

DHT11 Sensir Is connected to the pin Number D18 and VCC to 5v,GND to GND.

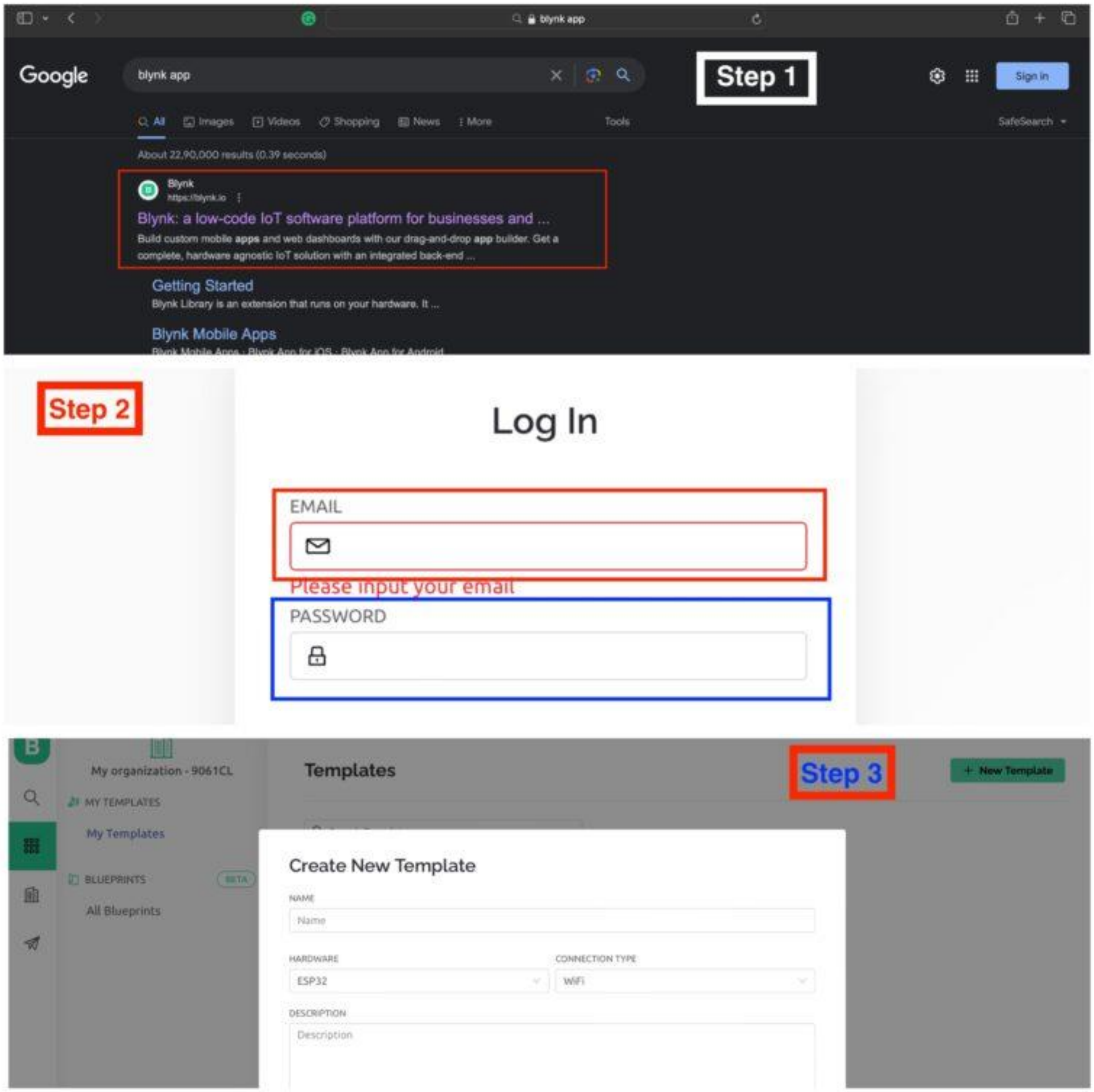
DS18B20 sensors is connected to the pin number D19 and VCC to 5v,GND to GND.

Blynk App Setup

Download the Blynk app from the Google Play Store or Apple App Store.

Create a new project in the Blynk app and obtain the authentication token.

Add widgets in the app, such as a value display for temperature readings.



Source Code

Blynk Library — Download

Upload Code To ESP32 Board

Connect the ESP32 board to the computer using a USB cable

Select the appropriate board and port in the Arduino IDE.

Upload the Arduino code to the ESP32 board.

```
#include <WiFi.h>

#include <LiquidCrystal_I2C.h>

#include <DHT.h>

#include <OneWire.h>

#include <DallasTemperature.h>

#include <BlynkSimpleEsp32.h>

Char auth[] = “YOUR_AUTH_TOKEN”;

Char ssid[] = “YOUR_WIFI_SSID”;

Char pass[] = “YOUR_WIFI_PASSWORD”;

LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
// DTH11 Sensor
#define DHTPIN 18
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
// DS18B20 Sensor
#define ONE_WIRE_BUS 19
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
#define VIRTUAL_PIN_TEMP
BLYNK_VIRTUAL_PIN_START
#define VIRTUAL_PIN_HUMIDITY
BLYNK_VIRTUAL_PIN_START + 1
Void setup() {
 Serial.begin(115200);
 Lcd.begin(16, 2);
 Lcd.setBacklight(LOW);
 Dht.begin();
 Sensors.begin();
```

```
  WiFi.begin(ssid, pass);
  While (WiFi.status() != WL_CONNECTED) {
    Delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");
  Blynk.begin(auth, ssid, pass);
}
Void loop() {
  Blynk.run();
  Float temperature = readDTH11Temperature();
  Float humidity = readDTH11Humidity();
  Float dsTemperature = readDS18B20Temperature();
  Lcd.clear();
  Lcd.setCursor(0, 0);
  Lcd.print("Temp: ");
  Lcd.print(temperature);
  Lcd.print(" C");
```

```
  Lcd.setCursor(0, 1);
  Lcd.print("Humidity: ");
  Lcd.print(humidity);
  Lcd.print("%");
  Blynk.virtualWrite(VIRTUAL_PIN_TEMP, temperature);
  Blynk.virtualWrite(VIRTUAL_PIN_HUMIDITY,
humidity);
  Delay(1000);
}
Float readDTH11Temperature() {
  Float temperature = dht.readTemperature();
  Return temperature;
}
Float readDTH11Humidity() {
  Float humidity = dht.readHumidity();
  Return humidity;
}
```

```
Float readDS18B20Temperature() {
  Sensors.requestTemperatures();
  Float temperature = sensors.getTempCByIndex(0);
  Return temperature;
}
```

Conclusion

The IoT-based temperature monitoring system using ESP32, DTH11, DS18B20 sensors, and a 16×2 LCD display with the Blynk app provides an efficient and convenient solution for remote temperature monitoring. The system demonstrates the integration of IoT devices and cloud-based applications to enable real-time data monitoring and visualization. This project highlights the practical application of IoT technology in home automation and monitoring systems.